

ALVIN EKELUND

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EDUCATION

Harvard University

M.S. in Data Science (in progress)

- MIT cross-registration in robotics and reinforcement learning (in progress).

Cambridge, MA

Aug 2026 - Feb 2028

Aalto University

B.Sc. Industrial Engineering and Management; CS minor

- GPA: 4.9/5.0 (Honors); LLM productivity thesis: 5/5. Coursework completed Dec 2025; NUS exchange, Spring 2026.

Espoo, Finland

Aug 2023 - Feb 2026

RESEARCH AND ENGINEERING PROJECTS

Walkthrough - Snap Spectacles [\[Demo video\]](#)

Nov 2025

- Part of a 5-person team behind an AR grocery assistant for Snap Spectacles: navigation, product information, and checkout.
- Team placed **Top 5 overall at Junction 2025** and **2nd in the Snap AR Challenge**.

Chain-of-Thought Workspace Ablation [\[Report\]](#)

2026

- Implemented Qwen3-4B ablations, two-stream decoding, and a grading harness across **3 benchmarks** (GSM8K, MATH-500, AIME) using PyTorch and Transformers.
- Ran a pre-registered study (~26 A100-hours): J-space ablation raised MATH-500 error by 8.0 percentage points (95% CI: 2.7-13.3); the norm-matched random control had no effect.
- Found that written reasoning alone did not restore MATH-500 accuracy after internal representations were disrupted.

Personal Brain - MCP Memory Server [\[Code\]](#)

2026

- Built an MCP memory server for cross-session agent retrieval, backed by a Python/SQLite knowledge graph with LLM ingestion, semantic search, and graph traversal.
- Added memory decay and interactive graph visualization to manage and inspect persistent agent context.

EXPERIENCE

Miracle Consulting Group

Software Engineer (promoted from intern)

Software Engineer Intern

Helsinki, Finland

Aug 2026 - Sep 2026

May 2026 - Aug 2026

- Raised satisfactory answers from **82% to 99.8%** (+17.8 percentage points) for a Fortune 500 client's Oracle document assistant by improving retrieval and answer safeguards.
- Reduced response time from **~75s to ~20s** per question in testing (**~73% lower latency**); drove user testing and client acceptance (Oracle APEX, ORDS).
- Built client AI/analytics applications and a shared design system; automated nightly document ingestion into the search index.
- Built an agent harness spanning **4 business functions**: ERP, CRM, HR, and finance.

Cadentia Technologies

Founder's Associate

Business Development Intern

Helsinki, Finland

Aug 2025 - May 2026

Feb 2025 - Aug 2025

- Helped secure the company's **first Fortune 500 customer** by building Graph-RAG demos in **React and Python**; turned customer knowledge bases into searchable, cited answers and iterated with the CEO for sales meetings.
- Automated market research and built an internal AI video-production pipeline.

Aalto University Department of Computer Science

Teaching Assistant

Espoo, Finland

Sep 2024 - Dec 2024

- Taught introductory programming, developed course materials, and mentored students in debugging.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, JavaScript, R, Scala, Swift

Frameworks/Tools: PyTorch, Transformers, MCP, Claude Code, Codex, SwiftUI, StoreKit 2, React, Docker, Git/GitHub Actions, Oracle APEX/ORDS, SQLite, Neo4j, Cloudflare Workers/D1

LEADERSHIP AND RECOGNITION

Leadership: Finnish Marines NCO course valedictorian (#1/200); Slush Team Lead; Treasurer, Aalto Triathlon Club

Awards: Hack & Roll main prize (~1,000 participants); Junction Top 5 / Snap AR 2nd (team awards); Dean's Scholarship (2024, 2025)

Languages: English (native), Swedish (native), Finnish (full working proficiency)

Availability: 20 hours/week during M.S.; Summer 2027 internships. Expected graduation: February 2028.